FC V2 — Rev A EVT1

Power

External regulated 5.2 V POWER1 input

USB-C data/debug only; VBUS isolated

On-board 5 V buck removed

Dual LDO rails (3.3V digital + 3.3V IMU)

Retained 9/12 V auxiliary regulator

NA226 retained; analog battery monitor disabled

File: power.kicad_sch

Sensors and Peripherals

ICM-42688P (primary IMU) — SPI

BMI270 (backup IMU) — SPI

BMP388 barometer — I²C

Magnetometer (LIS3MDL) — I²C

AT7456E OSD — SPI

W25Q128 flash memory (16 Mbit) — SPI

CAN transceiver (SN65HVD230) — FDCAN

File: Sensors.kicad_sch

MCU

MCU: STM32H743VIT6
ARM Cortex-M7 @ 480 MHz
2 MB Flash, 1 MB RAM
LQFP-100 Package

Hardware floating-point unit (FPU)

16-bit/32-bit timers with PWM

4× SPI, 4× I2C, 8× USART/UART

2× FDCAN, USB OTG FS/HS

3× 16-bit ADC (up to 36 channels)

DMA with DMAMUX

Built-in UART signal inversion

Operating voltage: 1.71V – 3.6V

File: MCU.kicad_sch

Connectors

8× motor outputs (PWM)

4× servo outputs (PWM)

GPS + compass port (UART + I²C)

RC receiver (SBUS / ELRS UART)

Telemetry UART

CAN bus (4-pin)

Airspeed (I²C)

VTX (analog + 9/12 V)

DJI FPV (UART + SBUS)

USB-C port

File: connectors.kicad_sch

Comments

Power

Signal Bidirectional

POWER/OPENESC REDESIGN IMPLEMENTED — BENCH VALIDATION REQUIRED
Derived from upstream commit 0a60da1 under CERN-OHL-S-2.0

Desan Vasu

Sheet: /

File: Flight Controller.kicad_sch

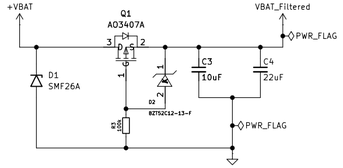
Title: Flight Controller V2

Size: User | Date: 2026-09-13 | Rev: A

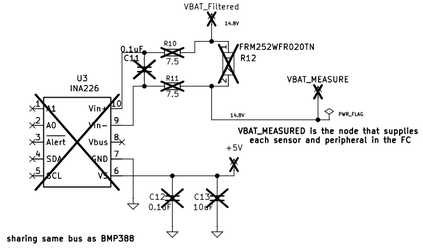
KiCad E.D.A. 10.0.6 | Id: 1/5

Power System

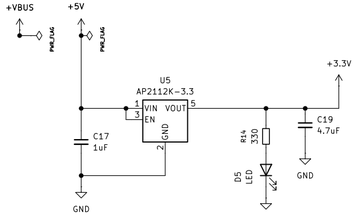
TVS and reverse polarity protection



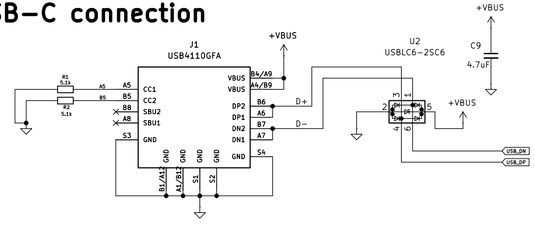
Current Sensing



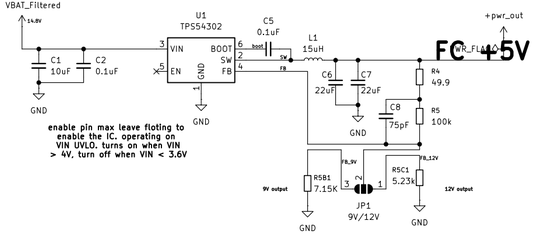
Main LDO



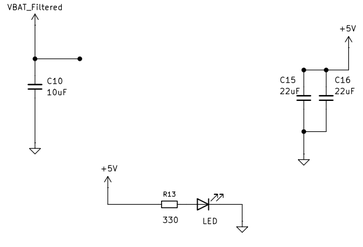
USB-C connection



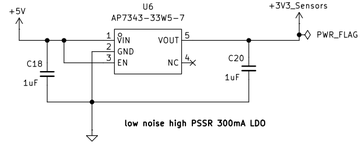
12V or 9V switching regulator



FC +5V buck removed — use external POWER1



IMU LDO



Comments

1) A saperate LDO for IMU to minimize the input noise per multiple firmwares advise.

2) Legacy INA226 / VBAT_MEASURED circuitry is retained but is not qualified as the EVT1 flight-battery failsafe source. Production firmware defaults BATT_MONITOR=0.

(pinouts and connection needs to be verified)

3) INA226 and BMP388 share the same I2C2 Bus
INA226 Address : 0X40 A0, A1 -> GND
BMP388 Address : 0X76 SDO -> GND

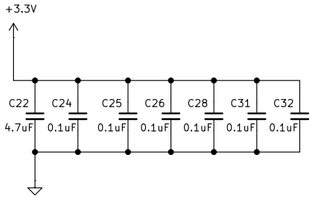
4) TPS54302 auxiliary 9/12 V regulator is retained unchanged. It is not the FC +5 V source and is excluded from the selected 3.0 A POWER1 load budget. External modules are not qualified here.

POWER/OPENESC REDESIGN IMPLEMENTED — BENCH VALIDATION REQUIRED
Derived from upstream commit 0a60da1 under CERN-OHL-S-2.0
Desan Vasu

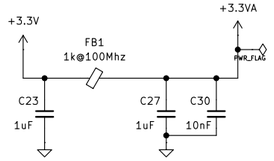
Sheet: /Power/		
File: power.kicad_sch		
Title: Flight Controller V2		
Size: User	Date: 2026-09-13	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 2/5

MCU

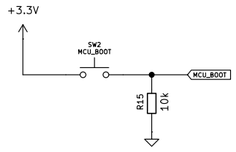
Decoupling and bulk Filtering



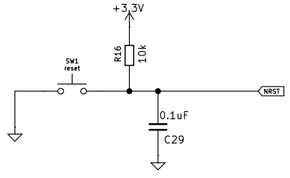
Analog Filter



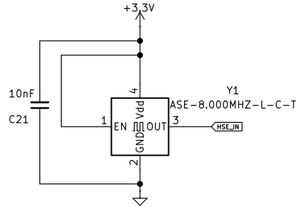
MCU_BOOT



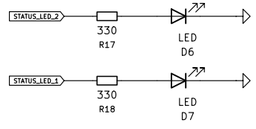
Reset

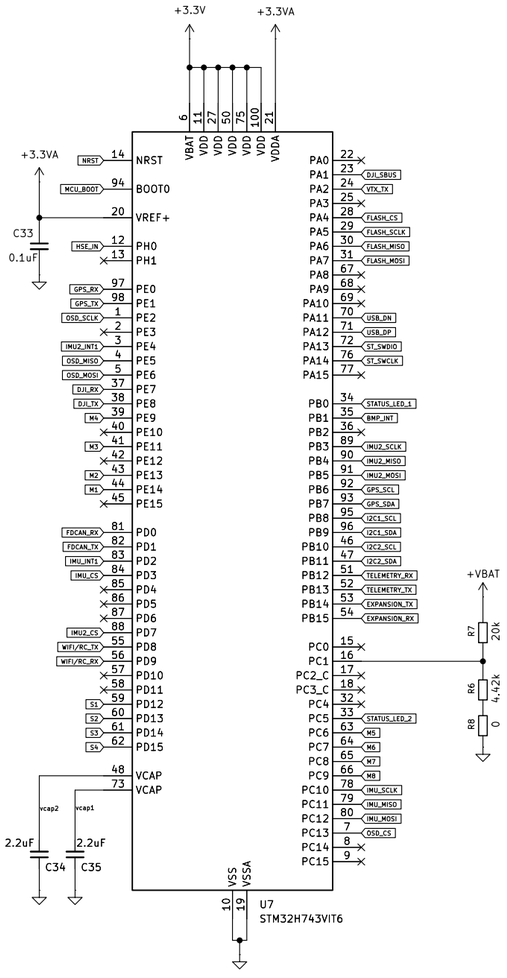


active 3.3 V CMOS oscillator



Status LED's





mouting holes 3.2mm each, applicable for M3 screws. maintin 30.5mm spaces in each direction in the layout

H1 MountingHole
H2 MountingHole
H3 MountingHole
H4 MountingHole

Comments

1) Instead of using a crystal resonator for external clock source, We will use CMOS, BYPASS clock oscillator. Safer to get right and less hassle/requirements

2) Motors PWM Pins were choosen to avoid DMA conflicts, motors utilize both Advanced timers TIM1 and TIM8 with 4 channels each.

Dont change the motors pin config

3) DJI_SBUS -> RC commands from DJI controller to FC.
100kbps
inverted, RX pin only
(UART4_RX)

IMU_INT1 -> GPIO_EXTI2
IMU2_INT -> GPIO_EXTI4
BMP388_INT -> GPIO_EXTI1
if you wish to change, make sure GPIO_EXT are diff for each sensor

4) On board Magnetometer and Airspeed sensor module will share the same I2C1 bus

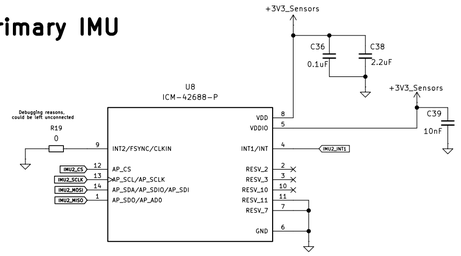
POWER/OPENESC REDESIGN IMPLEMENTED — BENCH VALIDATION REQUIRED
Derived from upstream commit 0a60da1 under CERN-OHL-S-2.0
Desan Vasu
Sheet: /MCU/
File: MCU.kicad_sch

Title: Flight Controller V2

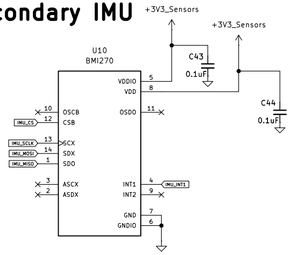
Size: User	Date: 2026-09-13	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 3/5

Sensors and Peripherals

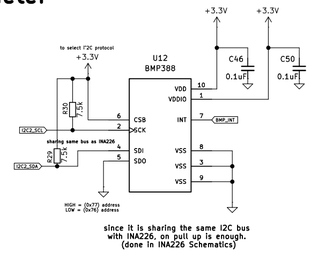
Primary IMU



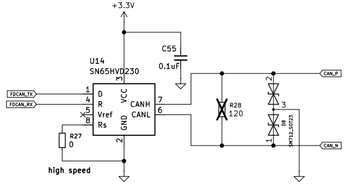
Secondary IMU



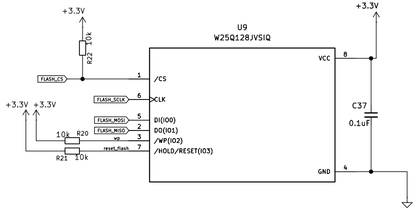
Barometer



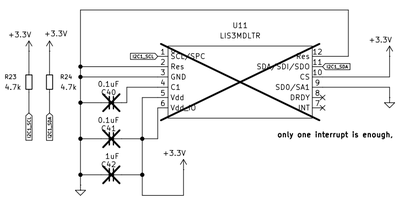
CAN Transceiver



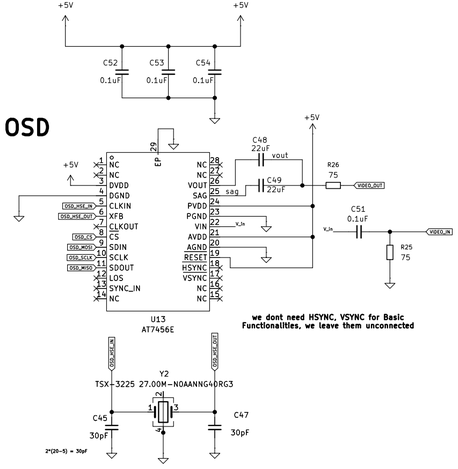
Flash Memory



Magnetometer



OSD





Comments

1) MCU HSE is supplied by the active 3.3 V CMOS ASE−8.000MHz oscillator. ArduPilot uses STM32_HSE_BYPASS; no passive−crystal load−capacitor calculation applies to Y1. Y2 remains the retained OSD crystal.

(Crystals weren't choosen. Oscillator modules for reliable clock generation)

2) A)CAN bus requires 120Ω termination resistors at both physical ends of the bus only. If the flight controller is positioned in the middle of the CAN chain, no termination should be installed. Populate R30 only if the flight controller is at one end of the bus.

B) 0Ω Resistor on RS for slope control, now its configured for High speed. If the wires are long then 10kΩ – 100kΩ should be used

3) On board magnetometer is cancelled and execluded from board. it is much more stable as an external module paired with GPS or individually.

POWER/OPENESC REDESIGN IMPLEMENTED – BENCH VALIDATION REQUIRED
Derived from upstream commit 0a60da1 under CERN−OHL−S−2.0

Desan Vasu

Sheet: /Sensors and Peripherals/
File: Sensors.kicad_sch

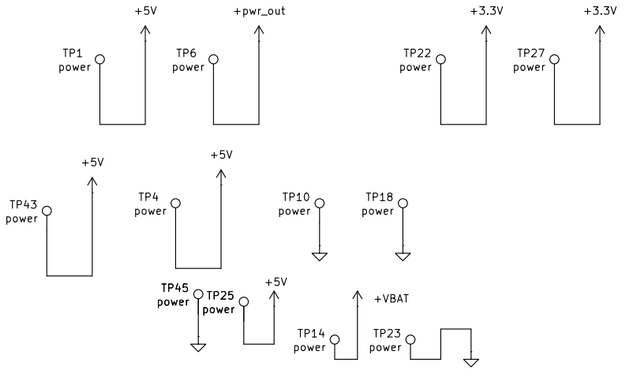
Title: Flight Controller V2

Size: User	Date: 2026-09-13	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 4/5

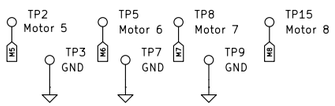
Connectors

Pads

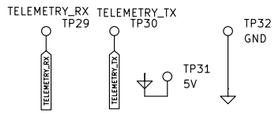
Power



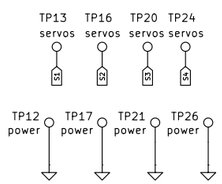
Motors PWM pins



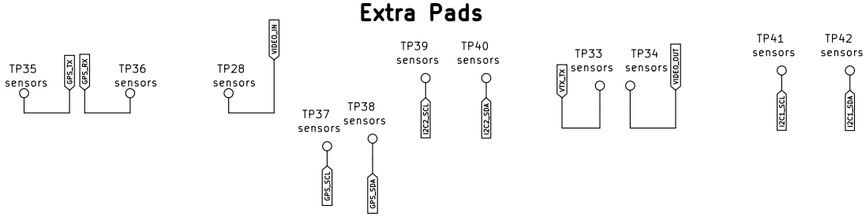
Telemetry



Servos

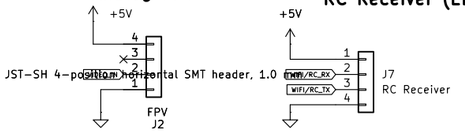


Extra Pads



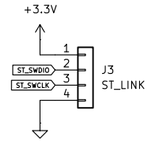
JST

Analog FPV

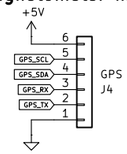


RC Receiver (ELRS)

ST-Link Debugger

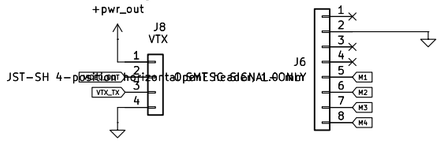


GPS + Magnetometer module

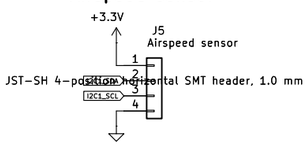


4 in 1 ESC

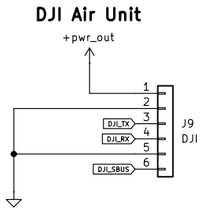
Analog VTX



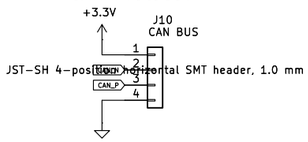
Airspeed sensor



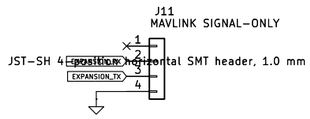
DJI Air Unit

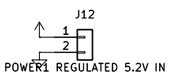


CAN Protocol



EXPANSION BOARD





Comments

1) DJI support, Direct plug and play for
   DJI O3 Air Unit
   DJI FPV Air Unit v1
   DJI FPV Vista / Caddx Vista
   DJI RunCam Link
   DJI Air Unit Lite (Vista variants)

2) J6: 8-pin JST-SH OpenESC signal-only.
   1 NC | 2 GND | 3 NC | 4 NC
   5 M1 | 6 M2 | 7 M3 | 8 M4
   No +VBAT, CURR, or +5V.

POWER/OPENESC REDESIGN IMPLEMENTED – BENCH VALIDATION REQUIRED
Derived from upstream commit 0a60da1 under CERN-OHL-S-2.0

Desan Vasu

Sheet: /Connectors/
File: connectors.kicad_sch

Title: Flight Controller V2

Size: User	Date: 2026-09-13	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 5/5